

Northwind Home: teaching the complaints agent what our words mean

Domain FDE Pre-Work Deep Track • Blue Data Consulting • Stage 4: Context

Where Day 3 left us

complaints_v3_measured is safe and measured. But when the team looked closely at what it did, three questions arrived that no rule or metric could answer:

- **Legal:** C-104 opted out of marketing email. Why was a plain apology about her own order blocked too?
- **Finance:** two refunds of 20 pounds on one complaint both went through. Is the 25 pound limit per refund, or per complaint?
- **Complaints team lead:** Margaret Doyle wrote again from a second email address, and the agent treated her as a new, routine customer.

None of these is a broken rule. Each one is a word nobody defined: **consent**, **refund limit**, **customer**. The agent read a one-line tool description, filled in the gaps with a confident guess, and nobody noticed because the data model looked fine.

Today's decision

What every word the agent relies on means, which thing it belongs to, and where that meaning is written so the agent actually reads it.

A diagram helps people. The agent only reads the instruction, the tool descriptions, the tool results and the conversation. If a meaning is not there, the agent does not know it.

The words that changed

Term	Day 3: implied	Day 4: defined	Belongs to
Customer	An email address on a complaint	One real person, who may have several accounts	Golden record
Vulnerable	A flag on one complaint	A circumstance of the person, applying to all their complaints	Customer
Consent	A list of opted-out channels	Permission per channel and per purpose: service or marketing	Customer
Refund limit	Each refund above 25 pounds	Total refunds on one complaint above 25 pounds	Complaint
Previous contacts	A number typed on the complaint	Earlier complaints from the same person, counted from the data	Customer

These definitions are assumptions waiting for sign-off by Legal and the complaints team lead. Whether a service apology may go to someone who opted out of marketing email is a legal question, not an engineering one.

What to watch

The terminal and scorecard work as on Day 3. New today is the ontology check, printed by the validation tests:

Result	Meaning
CONFORMS	The record fits every rule in the ontology
CONTRADICTION	The data breaks a rule, or disagrees with itself. Some are planted on purpose
GAP	A golden question cannot be answered by following links: an entity or relationship is missing

What was built

complaints_v4_context keeps the Day 2 rules and the Day 3 recorder. What changed is what the data means, and what the agent reads.

	complaints_v3_measured (Day 3)	complaints_v4_context (Day 4)
The data	Four complaints, with flags on each complaint	Customers, accounts, orders, complaints, refunds and consents, linked together
The words	Not written down	ontology.yaml: entities, relationships, definitions, synonyms and rules
What the agent reads	"Issue a refund." and other one-line descriptions	Tool descriptions and parameters written from the ontology, such as a purpose of service or marketing
The rules	Refund limit per refund; any message on an opted-out channel blocked	Refund limit per complaint; consent checked per channel and purpose; vulnerability read from the person
Proof	5 metric tests	Ontology validation: the data conforms, and 5 golden questions are answered by following links

Your task: 50 minutes, in pairs

CXM pairs. Work in usecase01_day4. Every command is in README-training-account.md. Record everything in Day4_Worksheets_UC01.docx. Do not change the ontology or the data; propose changes in ADR-4.

#	Part	Time	What you do
1	Define it cold	15 min	The opening questions, then your own entities, links and definitions for complaints. Before opening the code.
2	Validate it	10 min	Run the ontology tests. Find every planted contradiction, and note any your tests missed.
3	Run it	15 min	Run the same prompts on v3 and v4. Compare what each agent read, and what it did.
4	Prove it	10 min	Draft the golden set to 20 questions, write five questions for a data engineer, then ADR-4.

Seven questions

1. What does the agent actually read about a refund? Where exactly is it written, and what did it read on Day 3?
2. Vulnerability and consent moved from the complaint to the customer. Which Day 2 rule behaves differently because of that?
3. Is Margaret Doyle one customer or two? What links her accounts, and who decides?
4. C-104 opted out of marketing email. Should an apology by email be allowed? Who must sign that definition off?
5. Which planted contradictions did your tests find? Which did you only find by reading?
6. Pick a golden question the ontology cannot answer. What entity or link is missing, and what would you ask a data engineer?
7. Does "resolved" or "refund" mean something different to Complaints and to Billing? Single agent, router or orchestrator?

Bring to the debrief

The definition your pair disagreed on · the contradiction your tests missed · your ADR-4 choice, and the boundary that decided it.

All data is made up. Nothing changes in any real system. If the environment breaks, hand up.